Lecture 2 - 25/10

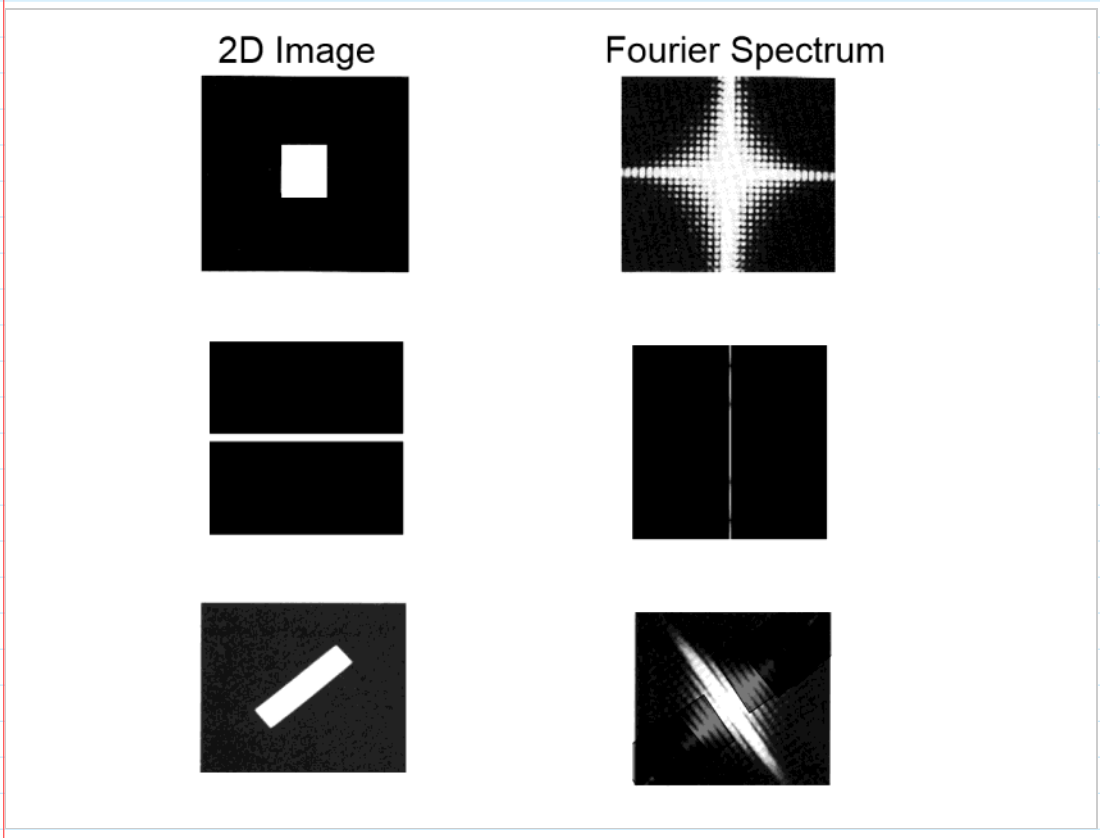
Friday, October 30, 2026 1:12 PM

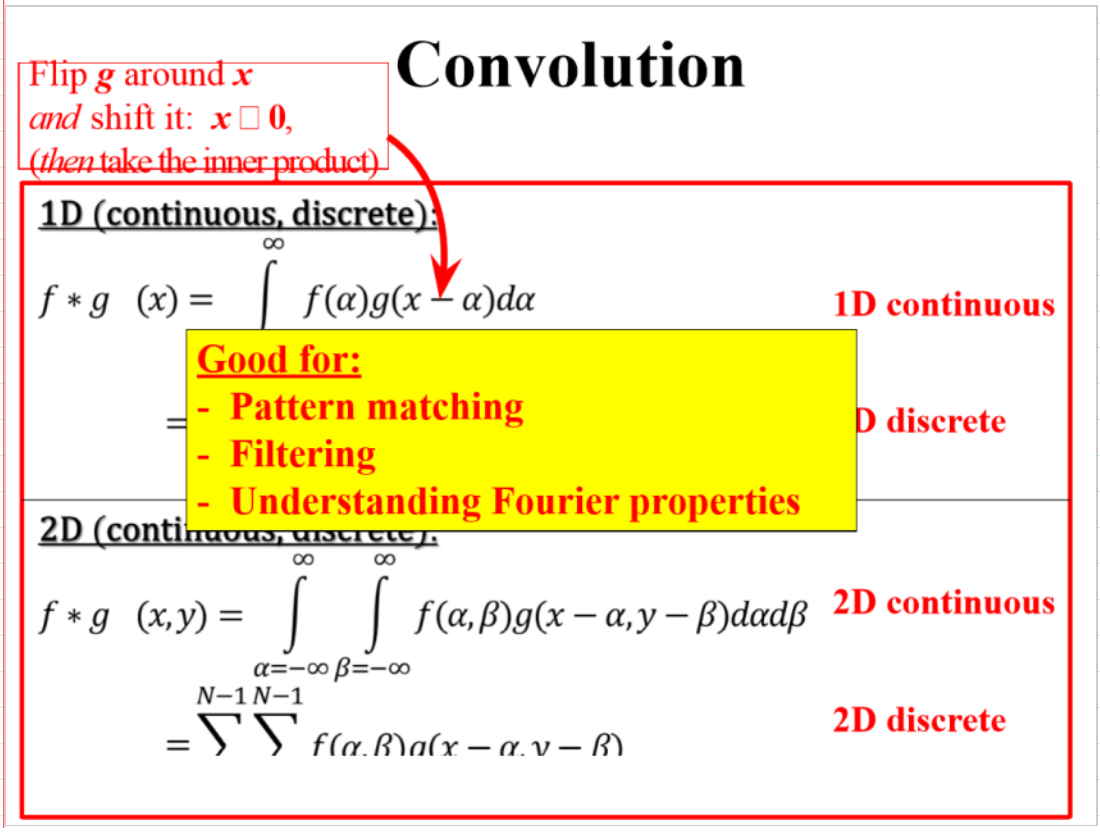
FourierII_slides

FourierII_slides

Convolution_pre_h...

https://docs.google.com/presentation/d/1YUv3qrtfDxR07KXgoe2oJHR3-aRCb2SiZr-yY_k8604/edit?slide=id.p1#slide=id.p1



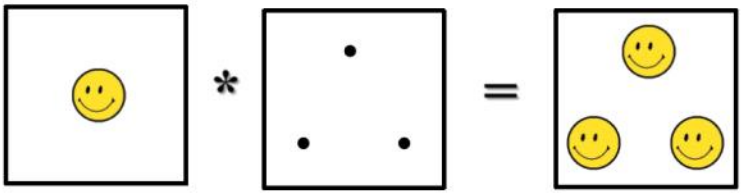


Examples

1) 

2) $f(x) * \delta(x) = f(x)$ **Unity**

3) $f(x) * \delta(x-d) = f(x-d)$ **Shift**

4)  **Replicate**

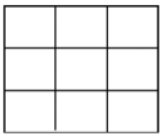
Convolution Properties

- Commutative:
  $f*g = g*f$
- Associative:
  $(f*g)*h = f*(g*h)$
- Homogeneous:
  $f*(\lambda g) = \lambda\, f*g$
- Additive (Distributive):
  $f*(g+h) = f*g+f*h$
- Shift-Invariant
  $f*g(x-x0,y-yo) = (f*g)\,(x-x0,y-yo)$

Proofs: Homework

Spatial Filtering Operations

3x3 filter

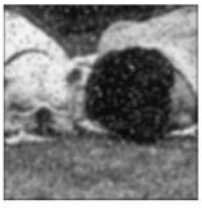


Example 3x3 filter

$$h(x,y) = f(x,y) * \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$h(x,y) = 1/9 \sum_{(n,m)} f(n,m)$$

Average of all pixels in the 3x3 neighborhood of (x,y)



3 X 3 Average



Salt & Pepper Noise



5 X 5 Average

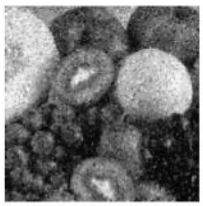
Local averaging: Removes noise but blurs edges



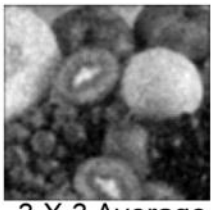
7 X 7 Average



Median



Salt & Pepper Noise



3 X 3 Average

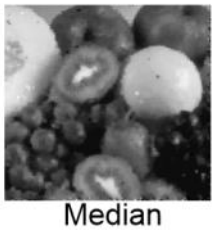


5 X 5 Average

Local averaging: Removes noise but blurs edges



7 X 7 Average



Median



$f(x,y) * \begin{bmatrix} 1 & -1 \end{bmatrix}$

$= f(x+1,y) - f(x,y)$

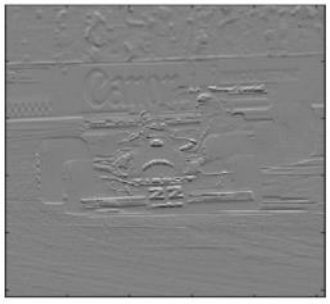
x derivative $\frac{\partial f}{\partial x}$



$f(x,y) * \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$= f(x,y+1) - f(x,y)$

y derivative $\frac{\partial f}{\partial y}$



A very simplistic "Edge Detector"

Gradient magnitude

$\|\nabla f\| = \sqrt{f_x^{\,2} + f_y^{\,2}}$



The Convolution Theorem

$$\Im\{f(x) * g(x)\} = \Im\{f(x)\} \cdot \Im\{g(x)\}$$

Proof: Homework

and similarly:

$$\Im\{f(x)g(x)\} = \Im\{f(x)\} * \Im\{g(x)\}$$

➔ Computational complexity of Discrete Convolution:
O(N *log*N)
Take f and g → FFT → $F \cdot G$ → FFT^{-1} → $f * g$

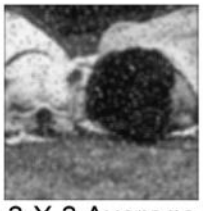
Going back to the "noise cleaning" example...



Salt & Pepper Noise

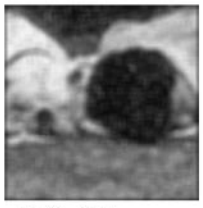
$$* \quad \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} =$$

rect



3 X 3 Average

Convolution with a ***rect*** □□ Multiplication with a ***sinc*** in the Fourier domain
= ***LPF*** (Low-Pass Filter)
□ **blurs the image**



5 X 5 Average

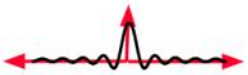
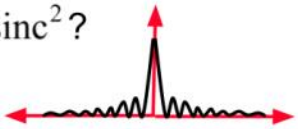


7 X 7 Average

Wider ***rect*** □□ Narrower ***sinc*** = Stronger ***LPF*** □ **Stronger blur**

Examples

What is the Fourier Transform of $f(x) = \text{sinc}^2$?

$f(x) = ($  $)^2 =$ 

$$\Im\{f(x)\} = \Im\{\text{sinc} \cdot \text{sinc}\}$$

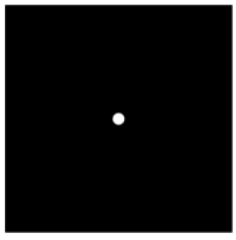
$$= \Im\{\text{sinc}\} * \Im\{\text{sinc}\}$$

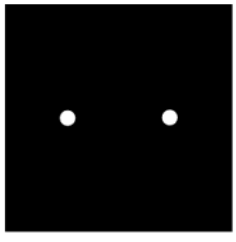
$$= rect * rect$$

$=$ 

$=$ 

Image Domain

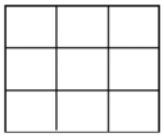




LPF in *Image domain* vs. *Fourier domain*

Computational complexity of convolution: O(N*log*N) via Fourier.
But for filter with a small support k □ O(kN) in the Image domain.

$k \times k$ rect



1) $f(x,y) *$ ↔ $F(u,v)\ sinc(u,v)$

Convenient
(small spatial support;
no cyclic assumption)

But not an ideal LPF...

2) $f(x,y) * sinc(x,y)$ ↔ $F(u,v)\ rect(u,v)$

An ideal LPF

3) $f(x,y) * gaussian(x,y)$ ↔ $F(u,v)\ Gaussian(u,v)$

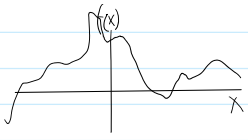
The Sampling Theorem

Developed on the board (also found in recording):

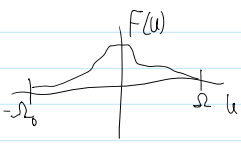
- **Nyquist frequency**
- **Aliasing**
- **Ideal interpolation**
- **Discretization induces a Cyclic function**



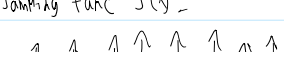
function



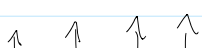
Fourier



sampling func $S(x) =$



$S(u)$



if F(u) is band limited -

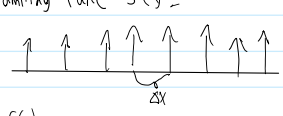
there is a certain frequency, above which all frequencies are 0, then we can sample it discretly

$\exists \Omega_0$ s.t. $\forall |u| > \Omega_0, F(u) = 0$

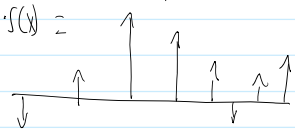
in 7a
they ask if we can sample f(x)=sin(ax)

from what we learnt, the sampling theorem, says we can sample it and reconstruct if F(u) is band limited
so what i actually need to do here, is show that F(u) is band limited?

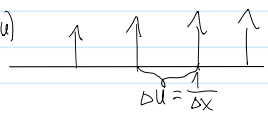
sampling func $s(x) =$



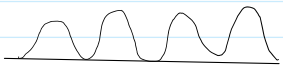
$f(x) \cdot s(x) =$



$S(u)$



$f(u) * s(u) =$



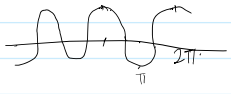
convolution with many dirac duplicates

$\exists \Omega_0$ s.t. $\forall |u| > \Omega_0, F(u) = 0$

lets look at cases

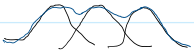
the "Nyquist frequency"

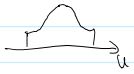
case 1: no overlap, if $\Delta u > 2\Omega_0 \Rightarrow \Delta x < \frac{1}{2\Omega_0}$



from here, take a LPF that captures one of the duplicates, thinks the original $F(u)$, then go to $f(x)$
so if we sample more densly than $2\Omega_0$, we can reconstruct

case 2: overlap



now if we take a LPF of this we get 
this causes aliasing

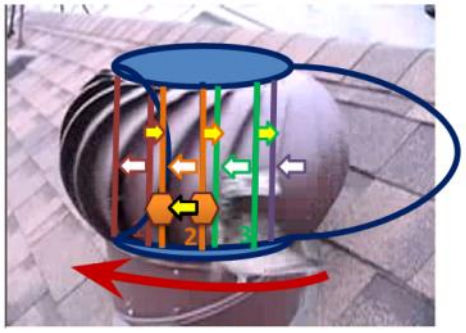
Examples of Temporal Aliasing

A "fast" dynamic event imaged by a "slow camera"

Motion aliasing



Examples of Temporal Aliasing



What's going on…?!

- The turbine rotates fast to the left.
- Each blade moves almost to the next blade position between frames.
  - So, we wrongly perceive rotation of the blades to the right
    - *TEMPORAL ALIASING!*

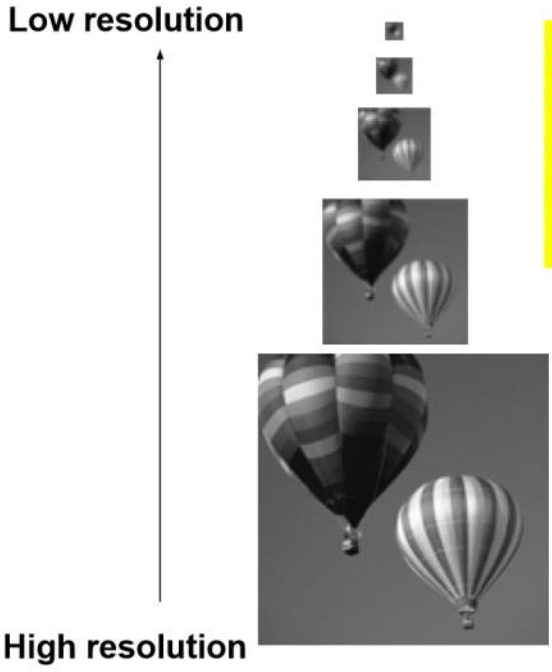
It happens because the temporal sampling rate (framerate) is too slow.

Multi-Scale Image Representation

- Gaussian pyramids
- Laplacian Pyramids
- Wavelet Pyramids



Image Pyramid



Good for:
- pattern matching
- motion analysis
- image compression
- other applications

Fast Pattern Matching

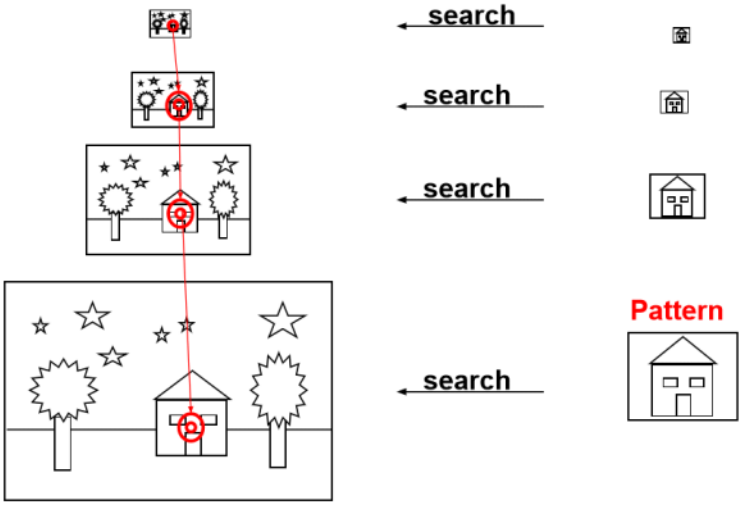
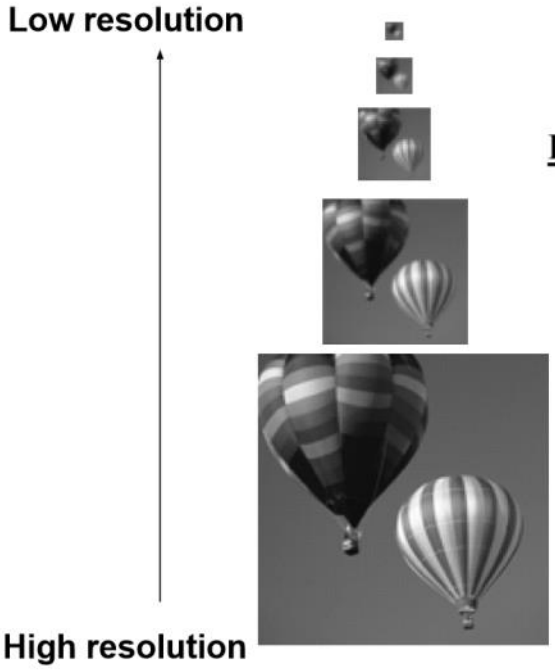


Image Pyramid

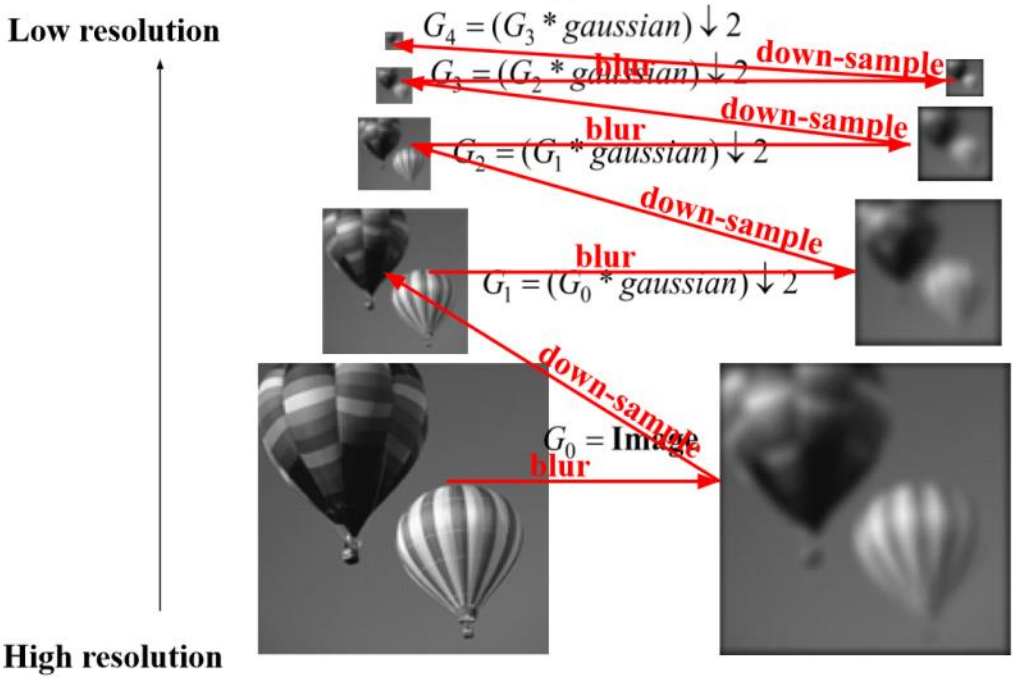


Does not consume much space

For an NxM image:

$$NM + \frac{1}{4}NM + \frac{1}{16}NM + \ldots$$
$$= \sum_l (\frac{1}{4})^l NM$$
$$\leq 1\frac{1}{3}\ NM$$

The Gaussian Pyramid



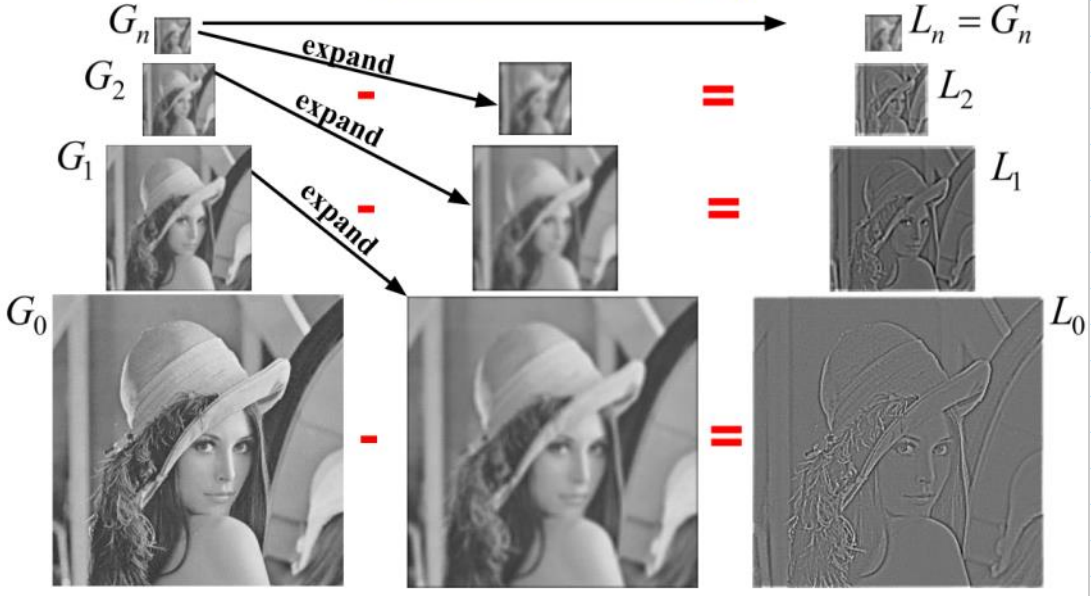
The Laplacian Pyramid

$L_i = G_i - \text{expand}(G_{i+1})$

$G_i = L_i + \text{expand}(G_{i+1})$

Useful for image compression

Gaussian Pyramid **Laplacian Pyramid**



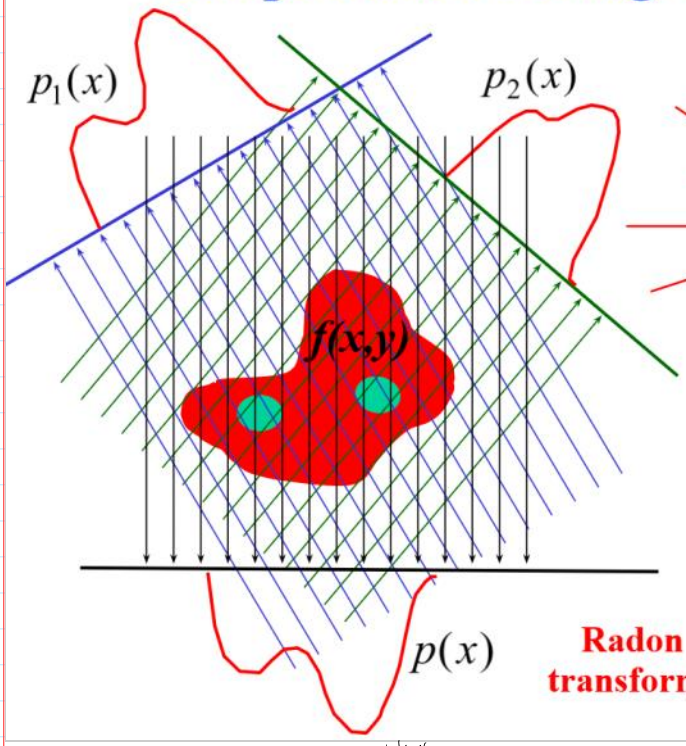
Laplacian ~ Difference of Gaussians

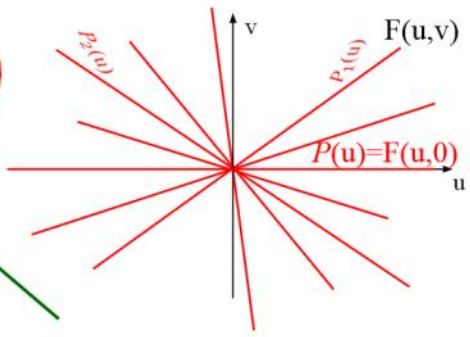


DOG = Difference of Gaussians

More details on Gaussian and Laplacian pyramids can be found in the paper by Burt and Adelson (link will appear on the website).

Computerized Tomography (CT)





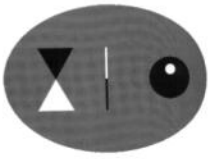
$$P(u) = F(u,0)$$

Radon transform: $$p(x) = \int_y f(x,y)dy$$

$$P(u) = \int_x p(x)\, e^{-2\pi i ux}\, dx = \int_x \int_y f(x,y)\, dy\, e^{-2\pi i ux}\, dx =$$

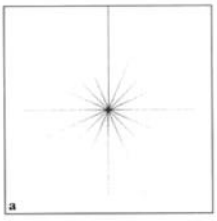
$$= \iint_{x,y} f(x,y)\, e^{-2\pi i (ux + 0y)}\, dy\, dx = F(u,0)$$

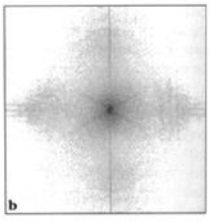
Computerized Tomography (CT)



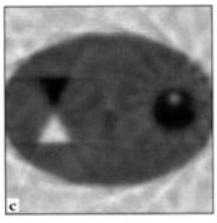
Original (simulated) 2D image

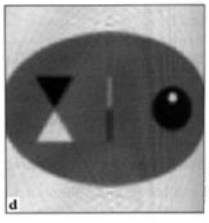
8 projections- Frequency Domain





120 projections- Frequency Domain





Reconstruction from 8 projections

Reconstruction from 120 projections

End of Lesson...

Exercise#1 -- will be posted on the website.
(Theoretical exercise: To be done and submitted individually)

To get course announcements:
please sign in to the course website! (Moodle)

Vision & AI Seminar (not for credit):
Thursdays at 12:15-13:15
(in person – in Ziskind-1)